Agent-Based Visualization: A flexible, modular and adaptive approach for visualizing complex data and simulation outputs

Arnaud GRIGNARD[1] and Alexis DROGOUL[1,2]

[1] UMI 209 UMMISCO IRD/UPMC, 32 Avenue Henri Varagnat, 93140 Bondy, France

[2]JEAI DREAM, IRD / University of Can Tho, 3/2 street, Ninh Kieu district, Can Tho city, Viet Nam

E-mail: [1]agrignard@gmail.com

Abstract

Data visualization is a modeling process that aims at representing rough data through visual abstractions. While they appear especially useful when datasets become too complex for the average user (such as in Big Data mining tasks), the techniques used are still lacking of the flexibility, modularity and adaptability required, especially when it comes to handle real-time visualization tasks, which is often the case when dealing with computer simulation outputs. The basis of our approach is to draw a parallel between the visualization of complex datasets and the modeling of complex systems, and to apply to visualization the agent-based paradigm, which has proved successful in the design of models of complex systems these last twenty years. The concepts and architecture have been implemented and experimented in the GAMA modeling and simulation platform [1].

Keyword: Visualization, Agent-Based Modeling

1. Data visualization and modeling

Data visualization, which is often associated with other data-mining techniques, can be used for descriptive, analytical or exploratory purposes [2]. In recent years, however, most of the existing visualization techniques have appeared to be inadequate with respect to the exploration and understanding of the large and complex datasets that compose the so-called Big Data domain. As a consequence, visualization is now often either neglected or considered as a secondary task, while it is of primary importance in many data-mining processes (providing an immediate feedback to users, or allowing them to intuitively explore datasets).

2. Agent-based visualization (ABV)

This inadequacy may be seen as the result of using techniques that have remained monolithic and centrally controlled, unable to handle the streams of data produced by heterogeneous data sources, sometimes in a dynamic fashion. The situation is reminiscent of that of the modeling of complex systems 20 years ago, i.e. before the advent and generalization of agent-based approaches to modeling. Agent-based visualization consists in applying the same recipe to visualization tasks: they are decomposed into discrete entities that are responsible of specialized visualization subtasks and can interact with other "visual agents". These agents have the possibility to learn about the collective/emergent patterns or behaviors present in the data they handle and to adapt their individual or collective aspect.

3. ABV methodology

We will show how to define an ABV process, from the reification of data into visual agents, the definition of their graphical representation to the application of abstraction and interaction rules to complete the visualization process.

ABV has proven, on various visualization tasks, to be flexible (in terms of defining different graphical representations and viewpoints on a particular dataset), modular (as agents can be added, removed or changed during the process) and adaptive (as the population of visual agents can reorganize to dynamically handle new data sources).

References

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